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Perfluoroalkyl Substances and Glycemic Status in Pregnant Danish Women: The Odense Child Cohort

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Abbreviations: BMI: Body mass index, CV: Coefficient of variation, DAG: Directed acyclic graph, IGT: Impaired glucose tolerance, IQR: Interquartile range, LOQ: Limit of Quantification, GDM: Gestational Diabetes Mellitus, GW: Gestational week, HOMA-IR: Homeostatic model of assessment of insulin resistance, HOMA-% β : Homeostatic model of assessment of β -cell function, IS_{OGTT}: Insulin sensitivity derived by oral glucose tolerance test, OCC: Odense Child Cohort, OGTT: Oral glucose tolerance test, PFASs: Perfluoroalkyl substances, PFDA: Perfluorodecanoic acid, PFHxS: Perfluorohexane sulfonic acid, PFOA: Perfluorooctanoic acid, PFOS: Perfluorooctane sulfonic acid, PFNA: Perfluorononanoic acid.

Abstract

Background: Perfluoroalkyl substances (PFASs) are persistent chemicals with suspected endocrine disrupting abilities applied in consumer products. PFASs have potentially modulating effects on glucose homeostasis. Insulin resistance prevails during third trimester of pregnancy, and this challenge of glucose homeostasis may reveal putative effects of PFAS concentrations on glycemic status.

Objective: To investigate associations between five serum PFASs and glucose-related outcomes in pregnant Danish women based on their risk of gestational diabetes mellitus (GDM).

Methods: In the prospective Odense Child Cohort serum concentrations of five PFASs - perfluorohexane sulfonic acid (PFHxS), perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA) and perfluorodecanoic acid (PFDA) - were measured at median gestational week (GW) 11 in pregnant women. An oral glucose tolerance test (OGTT) was performed at GW 28. The statistical analysis was conducted among 158 women with high GDM risk and 160 women with low GDM risk matched by gestational age. Multiple linear regression models were performed to estimate associations between PFAS concentrations and glucose, insulin, C-peptide, homeostatic model of assessment of insulin resistance (HOMA-IR) and beta cell function (HOMA-% β), and insulin sensitivity (Matsuda index) during the 2-h OGTT.

Results: In women with high risk for GDM, a two-fold increase in PFHxS concentration was significantly associated with increased fasting glucose, fasting insulin and HOMA-IR after adjusting for age, parity, educational level and pre-pregnancy BMI. Adjusting for the same confounders, a doubling in PFNA concentration was associated with higher fasting insulin and HOMA-% β .

In women with low GDM risk, no associations were found between PFAS concentrations and glucose-related outcomes.

Conclusion: PFHxS and PFNA concentrations were associated with impaired glycemic status in metabolically vulnerable pregnant women and might further enhance the risk of developing GDM.

Key words: Odense Child Cohort; Perfluoroalkyl substances; Glycemic status; Pregnancy; Oral glucose tolerance test

1. Introduction

Perfluoroalkyl substances (PFASs) are persistent chemicals with water, stain, and grease resistant properties, such as surface repellants for fabrics and food packaging (1). The two most prevalent PFAS exposure paths are from the diet and from inhalation of contaminated house dust in populations without high concentrations of PFASs in their drinking water (2). In human sera tested for PFASs, measurable concentrations are detected in the majority of samples (1, 3). PFASs bind with varying affinity to circulating proteins, mainly albumin (4). The mean serum half-lives of perfluorooctanoic acid (PFOA), perfluorooctane sulfonic acid (PFOS), and perfluorohexane sulfonic acid (PFHxS) were estimated in retired PFAS-production workers and reported to be 3.8, 5.4, and 8.5 years, respectively (5). PFOS concentration was associated with impaired glucose tolerance (IGT) and increased levels of serum insulin in rats (6) and adult human (7), however, effects of PFAS on glycemic status during pregnancy has not been thoroughly examined. Moreover, the reduced use of PFOS and PFOA over the last few decades has resulted in a shift to the use of other PFASs with unknown effects on human health (1, 8).

Insulin sensitivity is usually reduced during third trimester, and gestational diabetes mellitus (GDM) may develop in metabolically vulnerable women (9, 10). The third trimester represents a physiological window of opportunity to study possible effects of PFASs on glycemic status, as glucose homeostasis is already challenged. Four previous studies (11-14) in pregnant women looked at associations between PFAS concentrations and the diagnosis of GDM alone or GDM including IGT based on self-reported data (11) or extracted from medical records (12-14). However, glucose and insulin concentrations were not measured in any of the studies, and only one report included measurements of fasting glucose (15).

This study complements existing evidence by providing associations between serum concentrations of PFHxS, PFOS, PFOA, perfluorononanoic acid (PFNA), and perfluorodecanoic acid (PFDA) measured early in pregnancy and concentrations of glucose related-outcomes established by oral glucose tolerance test (OGTT) according to GDM risk during third trimester.

2. Materials and methods

2.1 Study population

The present study is part of Odense Child Cohort (OCC) (N=2,874), a longitudinal birth cohort conducted in Denmark, described in details elsewhere (16). Eligible women were residing in the Municipality of Odense, Region of Southern Denmark and were recruited at the

first antenatal visit (between 8-16 completed gestational weeks (GW)) between 2010 and 2012. Following enrollment, the women were requested to donate a blood sample for PFAS assessment at median GW 11 (N=649) and to complete a questionnaire on current health status. At about GW 28, a 2-h OGTT (including fasting, 1-h and 2-h glucose, insulin, and C-peptide) was performed in women with predefined risk factors for GDM (high GDM risk) (N=349) and in a random sample of women without those risk factors (low GDM risk) (N=270). Risk factors for GDM were BMI ≥ 27 kg/m², family history of diabetes mellitus, present multiple pregnancy, glucosuria during pregnancy, previous GDM or delivery of a macrosomic child (according to Danish screening criteria for GDM by Jensen DM, *et al.* (17)). The sample of women with low GDM risk were matched to the group of women with high GDM risk based on gestational age.

In this study, we excluded multiple pregnancy (N=56), women with ethnicity other than Danish (N=268) and GDM diagnosed before the third trimester (N=23) (according to the GDM diagnostic criteria by the European Association for the Study of Diabetes (18)). If a woman was pregnant more than once during the inclusion period (N=1), we only included the first pregnancy (Figure 1).

A total of 318 women had full data on PFAS concentrations, OGTT, and covariates, 158 and 160 of these subjects underwent an OGTT with a high GDM risk and a low GDM risk, respectively (Figure 1).

2.2. PFAS assessment

Assessment of serum PFAS concentrations included the following compounds: PFHxS, PFOS, PFOA, PFNA, and PFDA. Serum PFAS concentrations were assessed based on blood samples donated at inclusion (median (IQR): 11.3 (9.9; 14.3) weeks) in a subsample of 649 pregnant women. Of these, 200 were randomly selected and the remaining was selected based on availability of information from birth records, questionnaires and a clinical examination of the child at three months of age (19, 20). PFAS concentrations were estimated using online solid phase extraction followed by liquid chromatography and triple quadruple mass spectrometry (LC-MS/MS) at the Department of Environmental Medicine, University of Southern Denmark (19). The analyses were performed between September 2011 and September 2013. The within-batch coefficients of variation (CVs) were <3% and the between batch CVs were <5.2% (21). The Limit of Quantification (LOQ) was 0.03 ng/mL for all compounds. PFOS, PFOA, PFNA and PFDA were detectable in all samples in this study, but 0.9% of the partici-

pants had a PFHxS concentration below the LOQ that was reported as LOQ/2 (Supplementary Table 1).

2.3 Glycemic status

After an overnight fast, 2-h OGTT data (including plasma glucose, serum insulin, and serum C-peptide) were assessed after oral ingestion of 75 g anhydrous glucose dissolved in water. Assays were performed by the Clinical Biochemistry and Pharmacology Unit at Odense University Hospital, Denmark. Venous plasma glucose concentrations were measured by the hexokinase method (Architect, Abbott) with intra-assay CVs of 5.2-5.4% and inter-assay CVs of 1.2-1.7%. Serum insulin and serum C-peptide were analyzed by an electrochemiluminescence immunoassay (ECLIA, Cobas e411, Roche). Serum insulin intra-assay CVs were between 1.1-2.0% and inter-assay CVs were between 2.5-3.7% and for serum C-peptide intra-assay CVs were between 0.9-4.6% and inter-assay CVs were between 1.8-5.0%. The homeostatic model assessments (HOMA) of insulin resistance (HOMA-IR) and β -cell function (HOMA-% β) were calculated as described by Matthews *et al.* (22). Insulin sensitivity from the OGTT (IS_{OGTT}) was calculated by the equation described by Matsuda and DeFronzo (Matsuda index) (23).

2.4 Covariates

Anthropometric data (age, height and pre-pregnancy weight and body mass index (BMI)), parity, and smoking status were derived from hospital records using maternal social security numbers. Educational level (high school or less, high school + 1-4 years, high school + >4 years) was extracted from questionnaires, and missing data were retrieved from hospital records.

2.5 Ethical approval

The study was performed in accordance with the Helsinki Declaration II and approved by the Regional Ethical Review Committee (Project ID S-20090130) (16) and the Danish Data Protection Agency (j.no. 16/39607). All participants received written and oral information and provided their written consent for participation (16).

2.6 Statistical analysis

PFAS concentrations and glucose-related outcomes were not normally distributed and reported as medians. For analyses, the measures were transformed by the use of the natural loga-

rithm (ln) to obtain normally distributed residuals and variance homogeneity. Differences in PFAS concentrations, characteristics, and glucose-related outcomes between subjects with high GDM risk and low GDM risk were tested using unpaired t-test or Wilcoxon's rank sum test for continuous variables, and Chi squared test for categorical variables. Spearman correlations were assessed for each pair of PFAS.

Associations between PFAS concentrations and glucose-related outcomes were estimated using linear regression models. As both PFAS concentrations and glucose-related outcomes were ln-transformed, β -estimates were converted to express the percentage change in glucose-related outcome with a two-fold increase in PFAS concentrations. The assumptions underlying multiple linear regressions were checked and accepted. Augmented component-plus-residual plot was applied to identify non-linearities in the data. Normality and homoscedasticity of residuals in models were inspected graphically, and multicollinearity was checked formally. Analyses were performed separately in predefined groups of women with and without GDM risk factors, as they differed in glycemic status. Confounders and intermediate factors were identified using directed acyclic graphs (DAG (Supplementary Figure 1)) (24) based on *a priori* review of literature. High age (25), higher parity (26-28) and higher education (25, 26, 29) have been associated with lower PFAS concentrations. These covariates are also associated with insulin resistance during pregnancy (30-33). BMI may be an intermediate factor on the causal pathway between PFAS concentrations and glycemic status (34, 35). The adjusted model (Model 1) included age, parity (binary: 0; >0), and educational level (categorical), and pre-pregnancy BMI was additionally adjusted for (Model 2). Additionally, PFOS and PFOA concentrations were included in the regression model to investigate whether associations between PFHxS or PFNA and glucose-related outcomes changed. A significance level of 5% was used in all analyses. Data were analyzed using STATA/IC version 15.0 (StataCorp, College Station, TX, USA).

3. Results

The mean age was 29.9 (SD, ± 4.4) years, 61.3% were nulliparous, and 16.4% were in the highest educational group (Table 1). Four percent were smokers and the median pre-pregnancy BMI was 24.7 (IQR, 21.9-28.7) kg/m². Geometric mean and percentage above limit of quantification for the five PFASs are presented in Supplementary Table 1. Correlations between individual PFASs were weak to strong (between 0.27 and 0.73) (Supplementary Table 2). The included women did not differ from the excluded women according to age,

smoking status or educational level, but included women had a significant higher pre-pregnancy BMI and were more often nulliparous (data not shown).

3.1 Women with high GDM risk:

PFHxS and PFOA concentrations were significantly higher in women with pre-pregnancy BMI ≥ 27 kg/m², and PFHxS, PFOS, PFOA, PFNA and PFDA concentrations were significantly higher in nulliparous women (Supplementary table 3). Insulin sensitivity established by the OGTT (Matsuda index) was significantly lower and other glucose-related outcomes were significantly higher in women with high GDM risk compared to women with low GDM risk (Table 1). PFHxS and PFNA concentrations were significantly associated with impaired glycemic status (Table 2). After adjusting for age, parity, educational level and pre-pregnancy BMI, a two-fold increase in PFHxS concentration was significantly associated with increased fasting glucose, fasting insulin and insulin resistance (HOMA-IR). Adjusting for the same confounders, a doubling in PFNA concentration was associated with higher fasting insulin and β -cell function (HOMA-% β). Both PFHxS and PFNA concentrations were non-significant inverse associated with insulin sensitivity estimated by the Matsuda index. There were no evidence of associations between PFOS, PFOA, or PFDA concentrations and glucose-related outcomes (Table 2). Including PFOS or PFOA concentrations in the regression analyses did not weaken the observed significant associations between PFHxS or PFNA and glucose-related outcomes, actually associations were strengthened (data not shown).

Table 2, model 1, shows associations between PFAS concentrations and glucose-related outcomes after adjusting for age, parity and educational level, where BMI is treated as an intermediate factor on the causal pathway.

3.2 Women with low GDM risk:

No significant evidence of association was demonstrated between PFAS concentrations and markers of glucose-related outcome (Supplementary Table 4).

4. Discussion

Our study provides new information on associations between PFAS concentrations and glucose-related outcomes among pregnant women undergoing an OGTT stratified by GDM risk. In women with high risk of GDM, PFHxS concentration was associated with higher fasting glucose, fasting insulin, and HOMA-IR, and PFNA concentration was associated with in-

creased fasting insulin and HOMA-% β . Similarly, there were non-significant inverse associations between PFHxS and PFNA concentrations and insulin sensitivity estimated by the Matsuda index. PFOS, PFOA, and PFDA concentrations were not associated with glucose related outcomes. In contrast, no significant association was found between PFAS concentrations and glucose-related outcomes in women with low GDM risk.

Our mean concentration of PFNA was comparable to a Spanish cohort study of pregnant women with PFAS assessment at early pregnancy (13), PFHxS and PFOA concentrations were slightly lower and PFOS concentration was higher in our study. Our study findings need to be considered in light of four previous studies (11-13) that investigated associations between PFAS concentrations and diagnosis of IGT (12, 13) and GDM (11-14). In the first study (11), researchers observed a positive trend between pre-pregnancy PFNA concentration and self-reported GDM in U.S. women, and a higher PFOA concentration was associated with an increased risk of GDM. A second study (12) observed elevated odds ratios of IGT diagnosis in the second quartile compared to the first quartile of PFHxS concentration in Canadian pregnant women, but the authors found no significant associations between PFOS or PFOA concentrations and the diagnoses IGT or GDM. In the third study, researchers (13) observed positive associations between PFHxS or PFOS concentrations and risk of IGT diagnosis in pregnant women, but the odds ratio for PFHxS concentration was closer to null than PFOS concentration. However, the authors (13) reported a positive trend between PFHxS or PFOS concentrations and GDM diagnosis. In contrast, a Faroese study (14) found no clear association between PFOS, PFOA, PFHxS, PFNA, or PFDA concentrations and risk of GDM diagnosis in pregnant women. In our study, glycemic status was thoroughly studied biochemically at fasting and during OGTT (N=318), whereas the self-reported GDM (11) and chart reviewed IGT and GDM diagnoses (12-14) excluded any thorough assessment of glycemic status. We found no significant association between PFAS concentrations and glucose-related outcomes in women with low GDM risk undergoing OGTT. Researchers (15) reported inverse associations between PFOA, PFHxS, or PFNA concentrations and fasting glucose in pregnant women with no self-reported diabetes. These data suggest that the implications of PFAS concentrations on glucose homeostasis during pregnancy may depend on pre-conceptional metabolic vulnerability and that PFAS-associated pathologies may depend on the exposure profile, rather than the exact serum concentration of individual PFASs.

Serum concentrations of the individual PFASs are correlated and the temporal change in PFAS usage due to regulations (36) have affected the exposure profile and thus the ratio between individual PFASs in serum. This alteration may have influenced the ability to find

detectable effects of PFHxS and PFNA. Moreover, PFASs have been reported to be involved in several complex mechanistic pathways including but not limited to the peroxisome proliferator activated receptors (PPARs) (37). In vitro mouse and human cell studies (38-41) have demonstrated the ability of short- and long-chained PFASs to increase the activity of PPAR α and γ , nuclear transcription receptors, known to play essential roles in glucose and lipid metabolism (42). Activation of PPARs was shown to vary according to PFASs, and PFHxS and PFNA were found to be more potent than e.g. PFOS supporting our findings (39-41). Another potential mechanistic link may be an inhibition of the phosphorylation and activation of protein kinase B (AKT) as demonstrated in human hepatoma HepG2 cells for PFOS, but no other PFASs was investigated (43). The AKT is involved in cellular insulin signaling, and inhibition of AKT may result in hepatic insulin resistance and stimulation of gluconeogenesis (43). Collectively, the physiological net effect of PFASs on glycemic homeostasis in vivo is not clear and may differ depending on the potency of individual PFASs, concentration levels, and metabolic susceptibility.

A major strength of this study is the large group of population-based pregnant women with serum concentrations of PFASs. Moreover, we have assessments of glycemic status both in fasting state as well as during 2-h OGTT stratified according to pregnancies with high GDM risk as well low GDM risk. However, it was an exploratory study aiming to investigate trends across several glucose-related outcomes in relation to PFASs rather than associations with independent outcomes.

Women diagnosed with early GDM might have received nutritional counseling or even insulin injections in order to regulate glucose metabolism. As a result, this potential selection bias was dealt with by excluding women with early GDM diagnosed prior to GW 28. Relevant confounders were adjusted for, however, residual confounding by other predictors related to PFAS concentrations and glucose-related outcomes, such as dietary or lifestyle habits, cannot be ruled out.

Previous studies have suggested prenatal PFAS concentrations to be related with BMI (34, 35). As such, BMI may arguably be considered an intermediate factor on the causal pathway between PFAS and glycemic status. In our study, the total effect of PFASs on glycemic status was investigated both mediated through and not mediated through BMI. Moreover, the potential interaction between the five PFASs or co-exposure to other environmental chemicals on glucose homeostasis cannot be ruled out.

297

298 **5. Conclusion**

299 Increased serum PFHxS and PFNA concentrations were significantly associated with im-
300 paired glycemic status in pregnant Danish women at high risk for GDM. This finding sug-
301 gests that metabolically vulnerable pregnant women may be more susceptible to endocrine
302 disrupting chemicals, such as PFASs. However, our findings need replication and elucidation
303 of the mechanisms by which these chemicals interfere with glucose homeostasis.

304

305 **Declaration of interest, Funding and Acknowledgements**

306 **Declarations of interest:**

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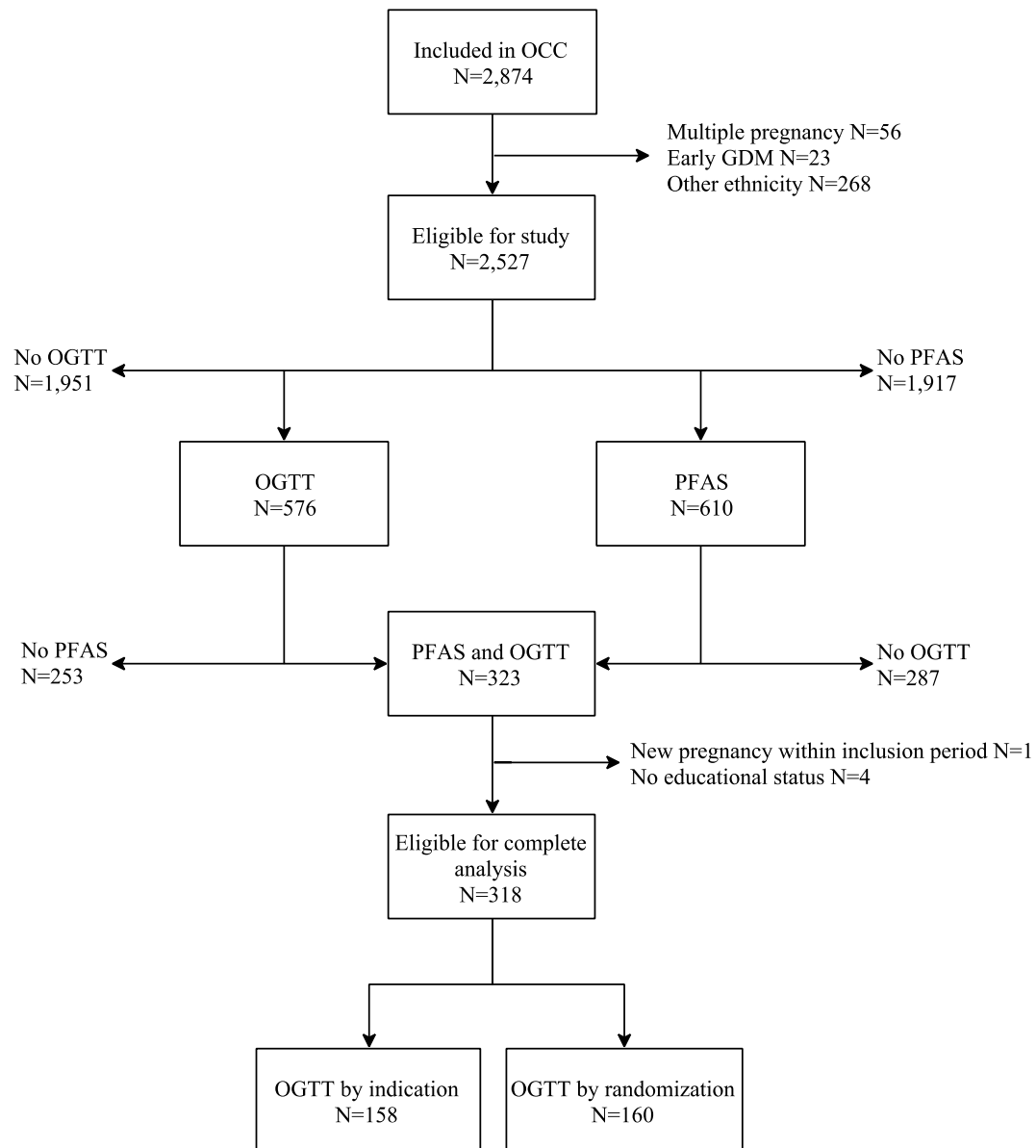
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Figure 1 Flowchart of the population in Odense Child Cohort (OCC) with serum PFAS concentrations and oral glucose tolerance test (OGTT)



1.5-column fitting image

Table 1 Characteristics, serum PFAS concentrations and glucose-related outcomes by OGTT groups (high GDM risk/low GDM risk)

Variables	All (N=318)	High GDM risk (N=158)	Low GDM risk (N=160)	P-value
Age, years	29.9 ($\pm$ 4.4)	30.3 ($\pm$ 4.7)	29.6 ($\pm$ 4.0)	0.15
Pre-pregnancy BMI, (kg/m ²)	24.7 (21.9-28.7)	28.5 (24.5-31.1)	22.7 (21.0-24.8)*	<0.001
Parity				
Nulliparous (%)	61.3	60.8	61.9	0.84
Primi+Multiparous (%)	38.7	39.2	38.1	
Smoking				
Yes (%)	4.4	5.7	3.1	0.26
No (%)	95.6	94.3	96.9	
Educational level				
High school or less (%)	34.0	33.1	33.8	0.92
High school +1-4 years (%)	49.6	48.7	50.6	
High school +>4 years (%)	16.4	17.1	15.6	
PFAS				
PFHxS (ng/mL) ^a	0.30 (0.08-0.60)	0.31 (0.08-0.66)	0.29 (0.07-0.60)	0.15
PFOS (ng/mL) ^a	8.31 (4.08-16.26)	8.37 (3.91-16.50)	8.15 (4.21-15.53)	0.64
PFOA (ng/mL) ^a	1.71 (0.69-4.19)	1.67 (0.67-3.80)	1.74 (0.71-4.79)	0.32
PFNA (ng/mL) ^a	0.66 (0.37-1.58)	0.65 (0.36-1.67)	0.69 (0.37-1.57)	0.82
PFDA (ng/mL) ^a	0.26 (0.15-0.53)	0.26 (0.15-0.51)	0.27 (0.15-0.62)	0.10
Fasting glucose (mmol/L)	5.1 (4.8-5.4)	5.2 (4.9-5.6)	5.0 (4.7-5.3)*	<0.001
Fasting insulin (pmol/L)	79 (51-112)	91 (64-129)	67 (45-92)*	<0.001
Fasting c-peptide (pmol/L)	766 (603-1006)	872 (684-1120)	695 (531-884)*	<0.001
2-h Glucose (mmol/L)	6.7 (5.7-7.7)	7.1 (5.9-8.0)	6.4 (5.5-7.2)*	<0.001
HOMA-IR	2.6 (1.6-3.8)	3.1 (2.0-4.5)	2.2 (1.4-3.1)*	<0.001
HOMA-% β	140 (106-187,7)	153 (115-213)	133 (101-168)*	<0.001
IS _{OGTT}	3.7 (2.6-5.7)	3.3 (2.2-4.7)	4.4 (3.1-6.7)*	<0.001

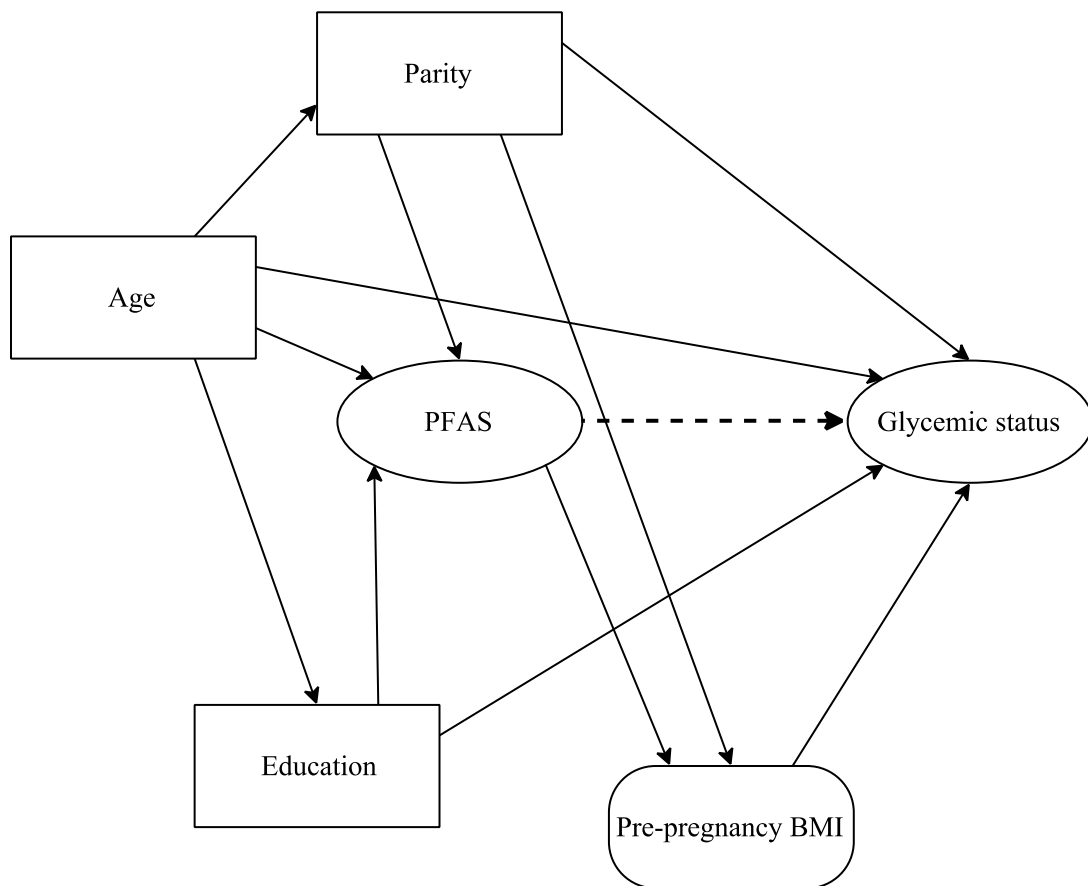
Data are presented as mean ($\pm$ standard deviation), median (interquartile range), ^a median (5th-95th percentile) and numbers (%). *P < 0.05, when comparing characteristics, serum PFAS concentrations and markers of glucose status by OGTT groups, t-test (normally distributed variables) or Wilcoxon's rank sum test (non-normally distributed variables) between groups, and Chi square test for categorical variables. IS_{OGTT}=Insulin sensitivity, Matsuda index

Table 2 Percentage change in glucose-related outcomes with a two-fold increase in serum PFAS concentrations in the high GDM risk group (N=158)

	Fasting Glucose (mmol/L)	Fasting insulin (pmol/L)	Fasting C-peptide (pmol/L)	2-h Glucose (mmol/L)	HOMA-IR	HOMA%β	IS _{OGTT}
	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)
PFHxS (ng/mL)							
Crude	2.4 (0.9; 3.9)*	10.9 (1.7; 20.9)*	7.7 (2.1; 13.7)*	4.2 (0.5; 8.0)*	13.5 (3.2; 24.9)*	3.1 (-4.2; 10.8)	-11.5 (-19.4; -2.8)*
Model 1 ^a	1.8 (0.3; 3.4)*	9.0 (-0.5; 19.5)	5.3 (-0.4; 11.4)	3.1 (-0.7; 7.2)	11.0 (0.3; 22.8)*	3.2 (-4.4; 11.4)	-8.0 (-16.6; 1.5)
Model 2 ^b	1.7 (0.2; 3.2)*	7.7 (0.1; 15.9)*	4.6 (-0.1; 9.6)	2.9 (-0.8; 6.8)	9.5 (1.0; 18.8)*	2.3 (-4.3; 9.4)	-7.1 (-14.8; 1.3)
PFOs (ng/mL)							
Crude	0.9 (-1.4; 3.3)	8.8 (-5.3; 24.9)	5.8 (-3.0; 15.4)	3.8 (-1.9; 9.9)	9.8 (-5.8; 27.9)	5.6 (-5.8; 18.3)	-8.8 (-21.5; 5.9)
Model 1 ^a	-0.1 (-2.5; 2.3)	2.4 (-11.3; 18.1)	0.4 (-8.0; 9.6)	2.8 (-3.1; 9.1)	2.3 (-12.7; 19.8)	2.7 (-8.8; 15.7)	-1.6 (-15.6; 14.7)
Model 2 ^b	-0.1 (-2.3; 2.2)	2.7 (-8.5; 15.2)	0.6 (-6.4; 8.2)	2.9 (-2.8; 8.9)	2.6 (-9.7; 16.6)	2.9 (-7.1; 14.1)	-1.9 (-14.3; 12.3)
PFOA (ng/mL)							
Crude	0.1 (-1.6; 1.8)	6.4 (-3.9; 17.8)	5.8 (-0.8; 12.7)	-0.4 (-4.5; 3.9)	6.5 (-4.9; 19.2)	5.7 (-2.8; 14.9)	-5.1 (-15.1; 6.0)
Model 1 ^a	-1.0 (-2.8; 0.9)	-3.8 (-11.2; 10.9)	0.3 (-6.3; 7.4)	-2.0 (-6.4; 2.6)	-1.8 (-13.1; 11.1)	1.9 (-7.1; 11.8)	3.1 (-8.4; 16.2)
Model 2 ^b	-1.3 (-3.0; 0.5)	-4.0 (-12.2; 5.0)	-1.5 (-6.9; 4.2)	-2.6 (-6.9; 1.8)	-5.2 (-14.2; 4.7)	-0.4 (-8.0; 8.0)	6.1 (-4.5; 17.8)
PFNA (ng/mL)							
Crude	1.3 (-0.9; 3.5)	19.1 (4.9; 35.2)*	11.9 (3.3; 21.1)*	0.8 (-4.4; 6.3)	20.7 (4.9; 38.8)*	14.7 (3.3; 27.4)*	-12.8 (-24.1; 0.2)
Model 1 ^a	0.4 (-1.9; 2.7)	16.0 (1.6; 32.6)*	8.5 (0.01; 17.8)*	-0.6 (-6.0; 5.2)	16.5 (0.5; 35.0)*	15.1 (3.1; 28.5)*	-8.1 (-20.5; 6.1)
Model 2 ^b	0.03 (-2.1; 2.2)	12.1 (0.7; 24.8)*	6.4 (-0.6; 13.9)	-1.3 (-6.5; 4.2)	12.2 (-0.5; 26.4)	12.4 (2.2; 23.7)*	-5.4 (-16.7; 7.5)
PFDA (ng/mL)							
Crude	-1.1 (-3.5; 1.3)	-2.7 (-15.9; 12.5)	-2.2 (-10.7; 7.7)	-2.9 (-8.6; 3.0)	-3.8 (-18.1; 13.0)	0.7 (-10.7; 13.5)	5.3 (-10.1; 23.2)
Model 1 ^a	-1.9 (-4.3; 0.5)	-6.8 (-19.3; 7.6)	-5.8 (-13.7; 2.8)	-4.5 (10.0; 1.3)	-8.6 (-22.0; 7.1)	-1.1 (-12.2; 11.5)	11.8 (-4.0; 30.3)
Model 2 ^b	-1.3 (-3.6; 1.0)	-0.2 (-11.2; 12.1)	-2.0 (-9.0; 5.4)	-3.3 (-8.7; 2.5)	-1.5 (-13.5; 12.1)	3.9 (-6.4; 15.2)	5.6 (-7.9; 21.1)

^a Adjusted for age, parity and educational level.^b Adjusted for age, parity, educational level and pre-pregnancy BMIIS_{OGTT}=Insulin sensitivity, Matsuda index.**P*< 0.05, in bold.

Supplementary Figure 1 Directed acyclic graph illustrating the causal pathway between pregnancy serum PFAS concentrations and glucose metabolism during pregnancy. The dashed line represents the exposure-outcome association to be tested, and solid lines illustrate the assumed association. Variables in rectangles represent potential confounders to be adjusted for, and variable in the rounded rectangle illustrates a causal intermediate factor.



2-column fitting image

Supplementary table 1 Serum PFAS concentrations during pregnancy

Perfluorinated compound	This study (N=318)		
	% < LOQ ^a	Median (IQR)	Geometric Mean (95 % CI)
Serum PFHxS (ng/mL)	0.9 %	0.30 (0.21-0.41)	0.27 (0.25; 0.29)
Serum PFOS (ng/mL)	0 %	8.31 (6.20-11.03)	8.23 (7.87; 8.60)
Serum PFOA (ng/mL)	0 %	1.71 (1.16-2.41)	1.68 (1.58; 1.79)
Serum PFNA (ng/mL)	0 %	0.66 (0.52-0.91)	0.70 (0.66; 0.73)
Serum PFDA (ng/mL)	0 %	0.26 (0.22-0.33)	0.27 (0.26; 0.28)

^a Values below limit of quantification (LOQ) were substituted by LOQ/2. CI: confidence interval, LOQ: Limit of Quantification, IQR: interquartile range

Supplementary table 2 Spearman correlations between five serum PFAS concentrations in pregnant women (n=318)

	PFHxS			
PFOS	0.56	PFOS		
PFOA	0.33	0.52	PFOA	
PFNA	0.47	0.54	0.63	PFNA
PFDA	0.27	0.53	0.56	0.73

Supplementary table 3 Median (5th-95th percentile) pregnancy serum concentrations of PFASs by characteristics in women with high GDM risk (n=158)

Variables	N (%)	PFHxS (ng/mL)	PFOS (ng/mL)	PFOA (ng/mL)	PFNA (ng/mL)	PFDA (ng/mL)
Median (5 th -95 th percentile)	158 (100)	0.31 (0.08; 0.66)	8.37 (3.91; 16.50)	1.67 (0.67; 3.80)	0.65 (0.36; 1.67)	0.26 (0.15; 0.51)
Age						
<25	17 (10.8)	0.26 (0.15; 2.28)	8.65 (5.11; 12.50)	1.84 (0.53; 3.76)	0.66 (0.26; 1.84)	0.24 (0.10; 0.38)
25-29	55 (34.8)	0.32 (0.04; 0.55)	7.94 (3.61; 16.81)	1.78 (0.79; 3.78)	0.72 (0.34; 1.93)	0.27 (0.16; 0.53)
30-34	55 (34.8)	0.30 (0.12; 0.53)	8.36 (4.46; 17.28)	1.67 (0.67; 3.89)	0.63 (0.43; 1.52)	0.26 (0.13; 0.51)
>35	31 (19.6)	0.39 (0.13; 0.97)	8.90 (3.56; 14.23)	1.34 (0.51; 3.70)	0.66 (0.37; 1.32)	0.24 (0.15; 0.52)
<i>p</i> ^a		0.11	0.87	0.64	0.80	0.82
Pre-pregnancy BMI, (kg/m ²)						
<27.0	56 (35.4)	0.28 (0.05; 0.70)*	7.74 (3.79; 16.46)	1.33 (0.51; 3.78)*	0.63 (0.34; 1.53)	0.28 (0.12; 0.48)
≥27.0	102 (64.6)	0.35 (0.15; 0.62)	8.67 (4.68; 16.53)	1.72 (0.80; 3.85)	0.72 (0.41; 1.84)	0.24 (0.16; 0.53)
<i>p</i> ^a		0.03	0.14	0.02	0.06	0.48
Family history of diabetes mellitus						
Yes	23 (14.6)	0.32 (0.20; 0.52)	7.94 (4.62; 14.91)	1.74 (0.94; 3.00)	0.67 (0.44; 1.43)	0.30 (0.17; 0.40)
No	90 (57.0)	0.34 (0.14; 0.70)	8.72 (3.91; 14.91)	1.44 (0.54; 3.52)	0.72 (0.36; 1.84)	0.24 (0.13; 0.48)
Missing	45 (28.4)					
<i>p</i> ^a		0.36	0.98	0.40	0.85	0.16
Parity						
Nulliparous	96 (60.8)	0.36 (0.09; 0.69)*	9.21 (4.83; 16.53)*	1.93 (0.83; 3.85)*	0.75 (0.43; 1.93)*	0.28 (0.16; 0.53)*
Primiparous+Multiparous	62 (39.2)	0.25 (0.08; 0.54)	7.02 (3.61; 14.39)	1.26 (0.54; 2.61)	0.56 (0.34; 0.96)	0.22 (0.14; 0.48)
<i>p</i> ^a		<0.001	<0.001	<0.001	<0.001	0.02
Smoking						
Yes	9 (5.7)	0.35 (0.04; 0.59)	8.24 (7.47; 12.01)	1.85 (1.06; 2.82)	0.65 (0.48; 2.03)	0.22 (0.16; 0.53)
No	149 (94.3)	0.31 (0.08; 0.66)	8.49 (3.91; 16.50)	1.65 (0.67; 3.80)	0.65 (0.36; 1.53)	0.26 (0.15; 0.48)
<i>p</i> ^a		0.60	0.66	0.59	0.70	0.34
Educational level						
High school or less	54 (34.2)	0.33 (0.08; 0.55)	8.29 (4.47; 16.53)	1.88 (0.71; 4.26)	0.63 (0.38; 2.08)	0.25 (0.12; 0.59)
High school +1-4 years	77 (48.7)	0.31 (0.05; 0.69)	8.85 (3.61; 15.30)	1.56 (0.57; 3.78)	0.66 (0.34; 1.53)	0.24 (0.14; 0.47)
High school +>4 years	27 (17.1)	0.36 (0.16; 0.70)	7.83 (3.79; 14.91)	1.47 (0.67; 3.20)	0.66 (0.46; 1.32)	0.29 (0.16; 0.48)
<i>p</i> ^a		0.25	0.83	0.23	0.43	0.44

^a Comparing PFAS median concentrations in categories of maternal characteristics and markers of glucose metabolism, Kruskal-Wallis test for more than two categories and Wilcoxon rank sum test for binary variables. *P<0.05, in bold

Supplementary table 4 Percentage change in glucose-related outcomes with a two-fold increase in serum PFAS concentrations in the low GDM risk group (N=160)

	Fasting Glucose (mmol/L)	Fasting insulin (pmol/L)	Fasting C-peptide (pmol/L)	2-h Glucose (mmol/L)	HOMA-IR	HOMA%β	IS _{OGTT}
	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)	% Change (95 % CI)
PFHxS (ng/mL)							
Crude	0.1 (-1.3; 1.5)	0.8 (-8.3; 10.9)	2.1 (-4.2; 8.9)	-0.3 (-3.7; 3.2)	0.9 (-9.1; 12.1)	0.4 (-6.9; 8.2)	1.0 (-7.7; 10.5)
Model 1 ^a	0.02 (-1.5; 1.5)	0.03 (-9.5; 10.6)	1.4 (-5.2; 8.5)	-0.3 (-3.9; 3.4)	0.04 (-10.4; 11.8)	-0.05 (-7.6; 8.1)	2.3 (-7.0; 12.4)
Model 2 ^b	0.2 (-1.3; 1.7)	2.6 (-6.7; 12.9)	3.0 (-3.5; 9.9)	0.2 (-3.4; 3.9)	2.8 (-7.5; 14.3)	1.9 (-5.5; 9.8)	0.1 (-8.6; 9.6)
PFOS (ng/mL)							
Crude	0.1 (-2.3; 2.6)	4.1 (-11.5; 22.5)	3.4 (-7.4; 15.5)	0.9 (-5.0; 7.1)	4.2 (-12.9; 24.8)	3.4 (-9.1; 17.6)	-7.7 (-20.9; 7.6)
Model 1 ^a	0.1 (-2.4; 2.7)	1.2 (-14.6; 19.8)	1.4 (-9.6; 13.7)	1.9 (-4.2; 8.3)	1.2 (-16.0; 22.1)	0.4 (-12.1; 14.6)	-6.3 (-20.1; 9.9)
Model 2 ^b	0.3 (-2.2; 2.8)	3.6 (-11.7; 21.6)	2.9 (-7.8; 15.0)	2.3 (-3.7; 8.8)	3.9 (-13.0; 24.0)	2.2 (-10.0; 15.9)	-8.2 (-21.2; 6.9)
PFOA (ng/mL)							
Crude	0.9 (-0.8; 2.6)	9.4 (-2.2; 22.4)	6.8 (-1.0; 15.2)	-0.9 (-4.9; 3.3)	10.4 (-2.5; 24.9)	5.9 (-3.1; 15.8)	-7.6 (-17.0; 2.7)
Model 1 ^a	1.2 (-0.8; 3.2)	9.0 (-4.4; 24.3)	6.2 (-2.8; 16.1)	0.1 (-4.8; 4.8)	10.3 (-4.6; 27.5)	4.6 (-5.6; 16.0)	-7.7 (-18.5; 4.4)
Model 2 ^b	1.2 (-0.7; 3.2)	9.4 (-3.3; 23.9)	6.4 (-2.2; 15.9)	-0.02 (-4.7; 4.8)	10.8 (-3.4; 27.1)	4.9 (-4.9; 15.7)	-8.0 (-18.3; 3.5)
PFNA (ng/mL)							
Crude	1.0 (-1.1; 3.2)	5.2 (-8.9; 21.4)	3.8 (-5.8; 14.4)	1.5 (-3.7; 7.0)	6.3 (-9.3; 24.5)	1.1 (-9.8; 13.3)	-8.2 (-19.8; 5.1)
Model 1 ^a	1.1 (-1.2; 3.4)	5.5 (-9.2; 22.7)	3.5 (-6.6; 14.6)	1.6 (-3.8; 7.3)	6.7 (-9.7; 26.0)	1.3 (-10.0; 14.0)	-8.4 (-20.5; 5.6)
Model 2 ^b	1.4 (-0.8; 3.7)	10.0 (-4.7; 26.9)	6.1 (-3.8; 17.0)	2.4 (-3.0; 8.2)	11.5 (-4.8; 30.7)	4.4 (-6.8; 17.0)	-11.6 (-22.9; 1.2)
PFDA (ng/mL)							
Crude	0.2 (-2.1; 2.6)	-2.1 (-16.2; 14.5)	-2.6 (-12.4; 8.3)	1.7 (-3.9; 7.7)	-1.9 (-17.4; 16.6)	-3.6 (-14.8; 9.0)	-5.8 (-18.7; 9.2)
Model 1 ^a	0.2 (-2.2; 2.7)	-2.0 (-16.6; 15.2)	-3.0 (-13.0; 8.2)	1.7 (-4.1; 7.9)	-1.8 (-17.8; 17.4)	-3.5 (-15.0; 9.5)	-5.9 (-19.2; 9.6)
Model 2 ^b	0.5 (-1.9; 2.9)	1.5 (-12.9; 18.3)	-1.0 (-10.8; 10.0)	2.4 (-3.4; 8.6)	2.0 (-13.9; 20.9)	-1.0 (-12.3; 11.7)	-8.7 (-21.1; 5.6)

^a Adjusted for age, parity and educational level.

^b Adjusted for age, parity, educational level and pre-pregnancy BMI

IS_{OGTT}=Insulin sensitivity, Matsuda index,

**P* < 0.05, in bold.